GSS AI Cost Modeling POC Requirement Document - Ver 1.0

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Revision Table

Version	Date	Description of Change	Author/Editor
1.0	12/16/2025	Initial Draft for Internal Review	Caio A Aggio, Dileep Sharma, Ashwini Shingote
1.1	1/5/2026	Added AI Module Overview	Ashwini Shingote
1.2	1/13/2026	Consolidated for external sharing	Anand Dhanaraju

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Project Objectives (Full Scope)

Use AI to Drive Savings with Actionable Insights, improve overall Cost Modeling process including the "Correlation" process keeping Cost Model tool as aPriori.

- Ingest current cost model data (cost model outputs, break-downs), and new offline models and match it to PO/Invoice data or quotation data
- Create tool to suggest opportunities: Comparison with respect to Target Cost, Supplier Quotes, and aPriori based Cost Model output
- **Potential future expansion** (not part of current revision): Create what-if modeling capabilities and scenario planning (e.g. what if there is change in sourcing location or customs tariff on Mexico).
    - Contribute to the overall vision of multi agent chatbot interface and architecture (Agentic Procurement Tool) (Refer to this slide).

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Business Challenges for Phase 1 (POC)

Business Problem: The Cost Modeling process is bottlenecked by extensive manual data extraction and workflow re-loops due to poor input quality by requester and lack of information for an effective correlation of quotes vs modeled cost. This limits the throughput of cost modeling and also results in fragmented data management that prevents effective cost optimization and quote correlation.

Workflow Delays/Re-loops

The request process to create a Cost Modelings is time-consuming and in many cases suffers from re-loops because critical inputs are missing from the user submission:

- Part mass, raw material, etc.
- Drawings not available.
- Version of drawing is incorrect.

Manual tasks

- Download drawings, read drawings, collect data to input in aPriori is very time consuming. (e.g., Wiring it needs to read the PDF and create a excel BOM out of it, also measure the small parts of the cables to calculate length of the cables).
- While the focus should be in optimizing the Cost Model and its Correlation to quotes, the Cost Model engineers spend a huge amount of time in manually extracting and manipulating data to make is useful for the Cost Model.

Data Management

- Cost breakdown and quotes most of the times are not available, requester ask for Cost Modeling just for a formal process, but are not really correlating it to the supplier quote.
- Cost Modeling outputs, quotes, cost breakdowns, correlation results are not well managed in a central system and database that can leverage AI and analytics. Currently is not easy to compare historical information, similar parts, etc.

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Phase Wise Scope

Phase 1 (POC)

Plastic Parts + Metal Parts:

1. A Classification module based on drawings (2D, 3D) and input templates.-
2. Reading PDF drawings & Input templates and capture specific attribute data, Go/No-Go decisions for prioritizing Cost Modeling requests on Dashboard

   Output: Cost modeling ready data to be put in aPriori.

3. Reading RM library (Only Raw Materials Whirlpool purchase) data from system on real time basis and capturing input for Cost modeling.
4. User uploads Cost breakdown from Quote in the *similar format as that of Cost modelling then AI will put it side by side and analyze.
5. Correlation Module will compare Cost Breakdown vs Cost Modeling output generated and provide recommendations to close the gaps

* Can be different formats, but AI wll have to be able to put it side by side with the Cost Modeling.

Phase 2 (To be refined based on Phase 1)

Electronics + Wiring*:

1. A Classification module based on drawings and input templates.
2. Reading PDF drawings & Input templates, Go/No-Go decisions for prioritizing Cost Modeling requests on Dashboard
3. Reading RM library data (wires and T2 component viz. Terminals , housing etc. from system on real time basis and capturing input for Cost modeling .
4. User upload Cost breakdown from Quote in the *similar format as that of Cost modelling then AI will put it side by side and analyze.
5. Correlation Module will compare Cost Breakdown vs Cost Modeling output generated

Electronics + Wiring: Need to evaluate current process that is being done in Excel with Macros (aPriori is not being used), there are 2 options to be considered:

Option 1 (preferred): Procure aPriori additional modules to work for Wiring/PCBAs. This will require to engage with aPriori technical team.

Option 2: Create the full process leveraging AI to even read drawings and perform the whole Cost Model in a more robust platform that will be connected with the Classification and Correlation Module.

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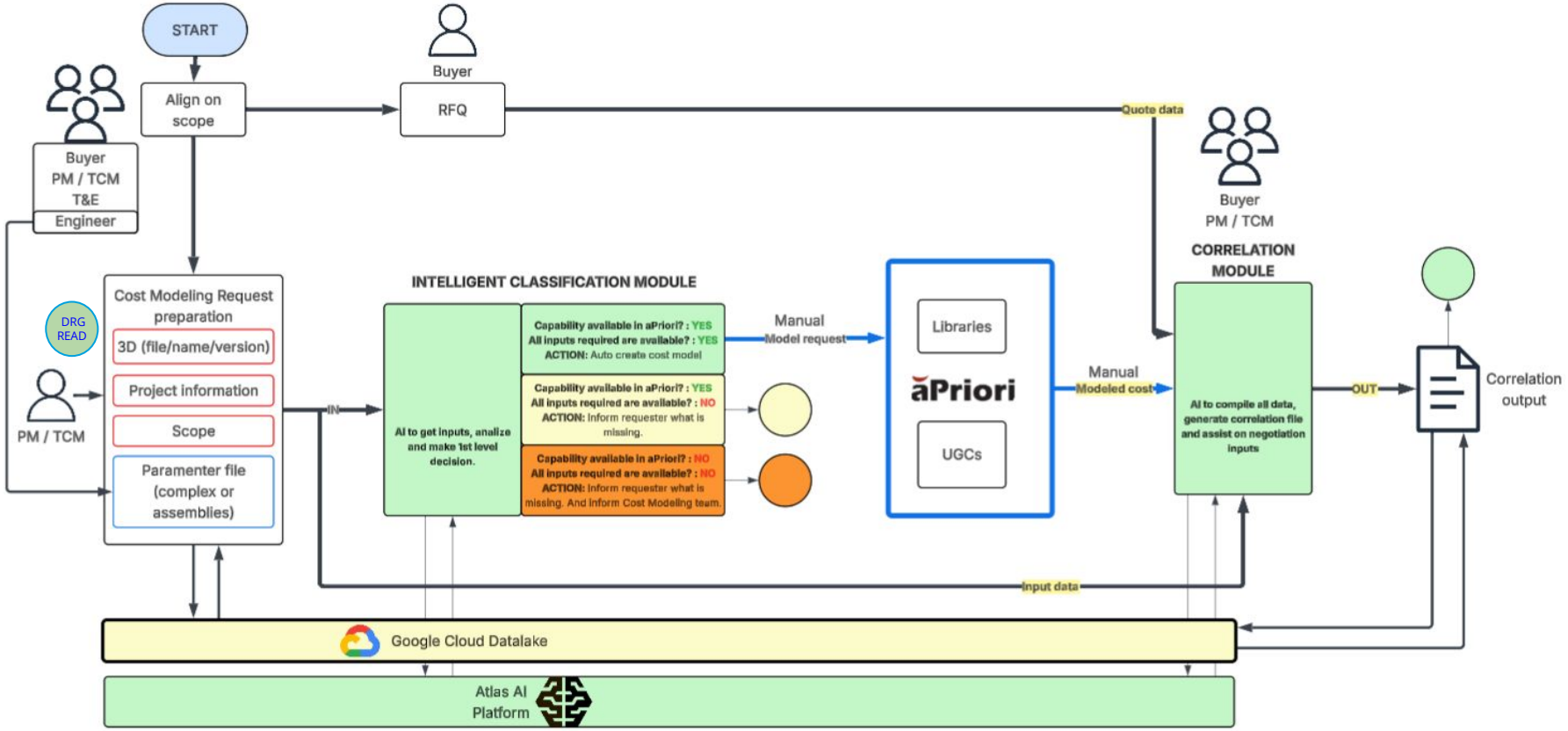
Fix the Diagram

Phase 1 (POC) Schematic Diagram - High Level



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Low Fidelity Wireframes Phase 1 (POC)

Welcome to AI Cost Intelligence

CREATE A NEW REQUEST

CORRELATION ANALYTICS

Status of Requests

Project Name	Part Numbers	Commodity Type	Status	Date of Completion	Link to see Details
Project Apollo	W123	Injection Molding (Plastics)	✅ Completed	Oct 24, 2023	LINK
Project Apollo	W124	Sheet Metal / Stamping	🟡 In Progress	Oct 24, 2026	LINK
Project Titan	W125	PCBA / Electronics	🔴 Pending	Dec 31, 2025	LINK
Project Titan	W126	Die Casting (Aluminum)	✅ Completed	Dec 31, 2024	LINK
Project Titan	W127	Machining (CNC)	❌ Canceled	N/A	LINK
NextGen EV	W128	Wiring Harness	🟡 In Progress	Dec 31, 2025	LINK
Legacy Retrofit	W129	Rubber Extrusion	🔴 Pending	Dec 31, 2025	LINK
Legacy Retrofit	W130	Rubber Extrusion	🔵 Correlated	Dec 31, 2025	LINK

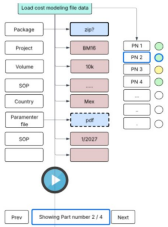
Export to Sheets

Legend:

- 🔵 **Correlated: Model results have been verified against actuals/quotes.**
- ✅ **Completed: The model is finished and ready for Correlation**
- 🟡 **In Progress: The Cost Modeling team is currently working on this.**
- 🔴 **Pending: Request sent but work has not yet started/assigned.**
- ❌ **Canceled: Request was withdrawn or no longer needed.**

NEW REQUEST PAGE

UPLOAD THE COST MODELING REQUEST FILE



INTELLIGENT CLASSIFICATION MODEL

Request match existing UGC and all inputs needed exist "Guess 40%"

Request match existing UGC, parameters are missing due to complex structure "Guess 40%"

Request does not match existing UGC "~20%"

CLASSIFICATION EXAMPLES

Sent for modeling

Missing parameters in the parameter file

No UGC available for the part submitted

Reasons of the classification
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris nulla ex, mollis vitae

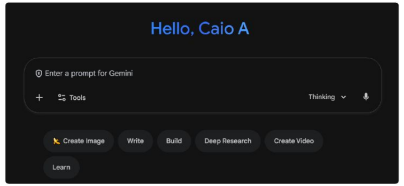
CORRELATION ANALYTICS

POC Required:

- Filter by project, part, requester, region, COO (Library), commodity type, volume, etc.
- History of Cost Breakdowns
- Comparison of Cost Breakdown vs Cost Modeling
- Insight to negotiation
- History of Quotes
- History of Modeled parts
- Chat-bot like interface

FULL Required:

- Part version history
- Material library changes history
- Early cost assessment history
- Alternative of similar Raw Materials to show opportunities
- GAP (Top offenders)
- Flags related to how many quotes (suppliers) were used



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PoC Phase 1 - Data Inputs

Process Inputs :

1. PDF drawings
2. Input Criteria sheet
3. 3D Model
4. Supplier quote template : Attach template here
5. Any other inputs

POC Inputs :

1. 50 plastic parts to Train the AI Model
2. 50 metal stamping parts to Train the AI Model

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Inputs

Files (sent as email attachments)

VMAX 2 NEXT GEN- Dryer Skirts W11673291 & W11673290-NAR-China_W11679887_harenss report

Cost Modeling Request

VMAX 2 NEXT GEN - Dryer Skirt- Cost Model Request- Sept 29 2023

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POC Phase 1 - Modules

Request Processing & Classification Module

Input :

1. PDF drawing (Windchill Global Vault)
2. Input Criteria sheet
3. 3D Model

Output :

1. Drawing parameters to Spreadsheet conversion
2. Gaps sheet between Drawing, 3D Model & Input criteria
3. Final evaluation of inputs as R/Y/G
4. Reject to requester with gaps & R/Y/G status
5. Request Status monitoring interface as a part of UI with various options/ drop downs (request no, platform category, Project name, evaluation status, submission date..
6. Data storage & retrieval for viewing past historic input data and assessment

Manual Costing Module

Input :

1. Drawing Parameter Spreadsheet
2. Gap Sheet
3. 3D Model
4. aPriori Costing Libraries

Output :

1. Cost spreadsheet for small to medium parts

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POC Phase 1 - Modules

Correlation Module

Input :

1. Drawing Parameter spreadsheet
2. aPriori Generated costing (manually input to the App)
3. Supplier Quote Spreadsheet (manually upload to the App)

Output :

1. Supplier quote to standard quote template conversion
2. Comparison sheet with supplier quote & aPriori generated costing
3. Recommendation output for differences
4. Correlation % for initial quote and after negotiation quote
5. User Interface for Correlation module
6. Data storage & retrieval for viewing past historic costing of similar parts

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Design Cost Process & Stakeholders

Step	Phase	Key Activity	Roles Involved
1	Target Setting	Defines the financial ceiling and provides cost allocations to the Design Team based on market value.	Target Costing Team (GSS)
2	Technical Design	Develops product architecture/3D CAD within the specific cost boundaries provided by Target Costing.	Design Engineers (Engg)
3	Cost Prediction	Executes Early Cost Assessment (ECA) using 3D CAD to predict technical costs with high correlation to final outcomes.	Cost Modeling Team (Manf/Quality)
4	Supplier Sourcing	Issues RFQs to the market and manages the intake of actual quotes from external vendors.	Initial Purchasing / Commodity Buyer (GSS)
5	Quoting	Provides binding commercial quotes based on the technical specifications received from Purchasing.	Suppliers
6	Gap Analysis	Reconciles the "Technical Model" vs. "Supplier Quote" vs. "Initial Target" to drive DTV (Design to Value) actions.	Cross-functional Team (GSS, Engg, Manufacturing)

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POC KPIs Current vs Expected (Metals+Plastic)

Modules	KPI	Current	Expected
Classification Module	Cost modelling input finalization **lead time** (request submission till acceptance once all inputs are provided)	Average of 18.6 days 70% of this days are due to not available data. **Average of 13 days**	1 Day for Green Status
Classification Module	**Reloops** **Caused due to missing information when model is requested**	25% of parts have at least 2 reloops	% of parts going to loops for POC .. After implementation we can align on target of 0 % reloop ? System to reject input if incomplete and inform user what is the issue.
Manual costing Module	**Extraction Accuracy Rate (%)** **(from dwg , pdf etc)**	Not available today as this is done manually	100%
Correlation Module	**No of Parts Correlated (%)**	No existing KPI - Not being measured	100% (for the CBD available parts)
Classification Module	**Library Management** Whirlpool Purchased Resins and Metals.	Manual	100 % Manual (Phase 1 POC)

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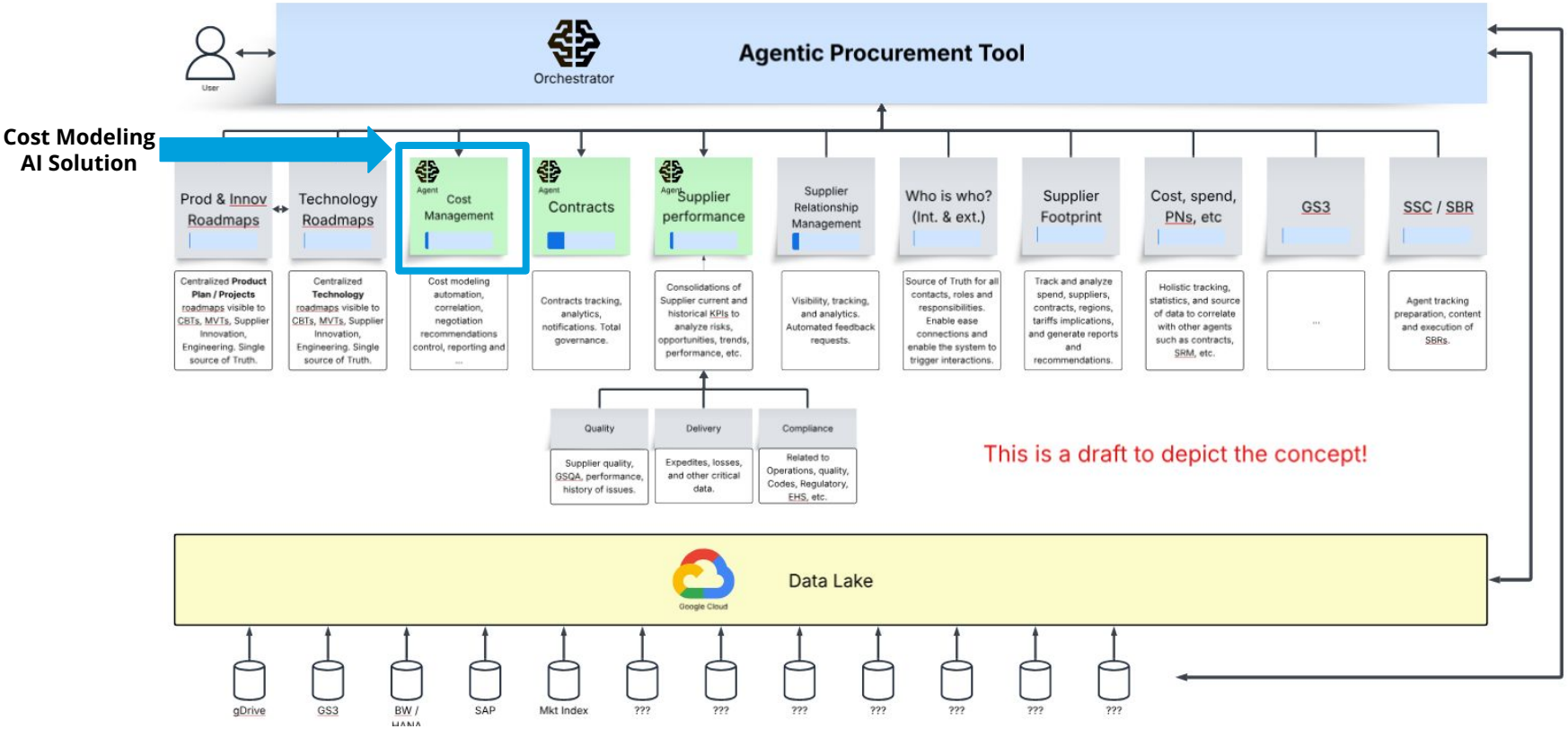
Backup

Agentic Procurement Tool - Long term vision (Draft)



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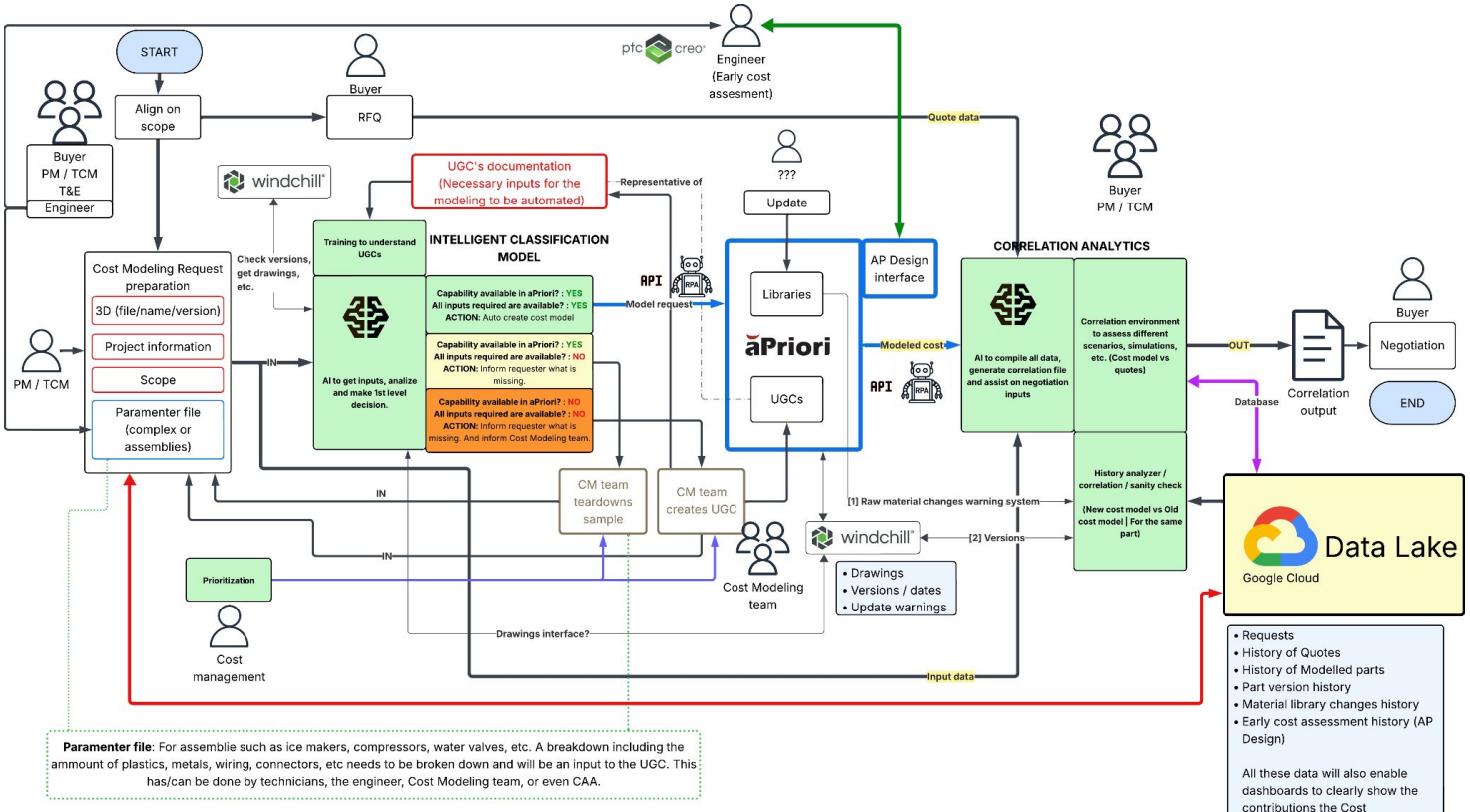
Complete Solution Schematic Diagram (Draft)



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Full Scope (Considering for all Cost Modeling Processes)

Functional Architecture & Process Flow

Module A: Request Ingestion & Preparation

The system must provide an interface for the Cost Modeling Request Preparation.

Inputs: The system must ingest 3D files (name/version), Project Information, Scope, and Parameter Files (for complex assemblies like ice makers/compressors).

Integration: Direct integration with Windchill (PLM) to fetch drawings, check versions, and validate data availability.

Prioritization: A mechanism for Cost Management to prioritize incoming requests.

Module B: Intelligent Classification Model

An AI logic layer required to analyze incoming requests and make 1st-level decisions:

Logic Check: The model must assess if the required capability and inputs exist within the aPriori libraries.

Automated Workflow (Green Path): If inputs/capabilities are present, the system automatically triggers the cost model creation.

Exception Handling (Orange/Red Path):

If inputs are missing: Automatically inform the requester.

If capability is missing: Route to the Cost Modeling Team to create the necessary User Guided Costing (UGC) or teardown samples.

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Full Scope (Considering for all Cost Modeling Processes)

Functional Architecture & Process Flow

Module C: Execution & Modeling (aPriori Integration)

API Orchestration: The solution must utilize APIs to drive aPriori.

Library Management: Integration with current libraries and UGCs.

Output: Generation of the "Modeled Cost."

Module D: Correlation Analytics

This is the core analytics engine that compares the Modeled Cost against Supplier Quote Data.

Data Compilation: AI compilation of all data points to generate a correlation analysis.

Scenario Planning: Environment to assess different scenarios and simulations to assist Buyers in negotiation.

Sanity Checks: Automated "History Analyzer" to compare New Cost Models vs. Old Cost Models for the same part number.

Ability to upload different formats of Cost Breakdown from Suppliers

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Full Scope (Considering for all Cost Modeling Processes)

Data & Analytics Requirements

The solution must feed into a Google Cloud Data Lake to power dashboards, historical analysis and Chat-bot like interface. The specific analytics features required are:

- Part Version History: Tracking cost evolution across different design versions.
- Material Library Changes History: detailed logs of how material cost inputs have changed over time.
- Early Cost Assessment History: Tracking estimates made by engineers (via AP Design) during early phases vs. final models.
- Alternative Raw Material Opportunities: AI-driven suggestions for similar raw materials that could offer cost savings.
- GAP Analysis (Top Offenders): Automated identification of parts with the largest delta between Whirlpool Cost Model and Supplier Quote.
- Supplier Quote Density: Flags indicating how many quotes (suppliers) were utilized for a specific comparison (to measure competitive tension).
- Filter by project, part, requester, region, COO (Library), commodity type, volume, etc.
- History of Cost Breakdowns
- Comparison of Cost Breakdown vs Cost Modeling
- Insight to negotiation
- History of Quotes
- History of Modeled parts

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Full Scope (Considering for all Cost Modeling Processes)

Technical Integrations

PLM: PTC Windchill (fetching drawings, versions, dates).

CAD: PTC Creo.

Costing Software: aPriori (utilizing Libraries, UGCs, and APIs).

Cloud Infrastructure: Google Cloud Platform (Data Lake storage and analytics computation).

Integrate with an Orchestrator Procurement Agent (*Not Developed yet*)

User Roles & Access Control

The system must support specific workflows/access for the following:

- Buyer
- Project Manager
- Target Costing Manager
- Cost Modeling

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Data Factor Map

Cost Modeling Reporting and Analytics Optimization

Product Design Data

- Product research
- Design, prototype data
- Engg. costs - CAD designs, drafting etc
- Tooling costs

Production costs data

- Direct Materials cost
- Direct labor costs
- Indirect materials cost
- Utilities
- Rent/lease costs
- Maintenance costs

Master Data

- Raw Material cost data
- Parts cost data
- Bill of Material
- CAD drawings

Supplier Data

- Location
- Commodity codes
- Volume data
- Supplier quotes

ERP system Dat

- SAP
- BW
- HANA DB
- Creo
- Apriori

Other data sets

- Buyer data sets
-

Historical data

- Production data
- Costs figues
- Trends- forecasts

External Source

- Supplier quotes
- Industry benchmarks
- Macroeconomic factors
- Technology advancements - automation etc

Assumption: The factor map has been created based on our understanding on all potential factors. We will modify factors based on data availability.

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Whirlpool KitchenAid JENNAIR MAYTAG AMANA

BRASTEMP Consul insinkerator